

Epidemiology of multiple sclerosis in US veterans. VIII. Long-term survival after onset of multiple sclerosis.

PMID: 10908197 | Indexed in PubMed

Survival to 1996 was analysed for nearly 2500 veterans of World War II who were rated as 'service-connected' for multiple sclerosis as of 1956 by the then Veterans Administration. Survival from onset was defined for all white women and black men, and a random sample of white men. Median survival times from onset were 43 years (white females), 30 years (black males) and 34 years (white males). Crude 50-year survival rates were 31.5% (white females), 21.5% (black males) and 16.6% (white males), but only the white females and white males were significantly different. A proportional hazard analysis was used to identify risk factors for mortality from multiple sclerosis onset year. Significant risk factors included male sex (risk ratio: 1.57), older age at onset (risk ratio: 1.05 per year) and high socioeconomic status (risk ratio: 1.05 per socioeconomic status category). There were no statistically significant differences in survival following multiple sclerosis onset by race or latitude of place of entry into military service, both significant risk factors associated with the development of multiple sclerosis. Standardized mortality ratios utilizing national US data (for 1956-96) showed a marked excess for all three race-sex groups of multiple sclerosis cases, with little difference among them, but with a decreasing excess over time. Relative survival rates, used to compare the survival of multiple sclerosis cases with that of other military veterans, did not differ significantly by sex-race group, nor by latitude of place of entry into military service, but did differ significantly by socioeconomic class. The lack of difference in male and female relative survival rates suggests that the significant difference in survival between male and female multiple sclerosis cases is, at least in part, a result of sex per se and not the disease.

Excess Mortality in Patients with Multiple Sclerosis Starts at 20 Years from Clinical Onset: Data from a Large-Scale French Observational Study.

PMID: 26148099 | Indexed in PubMed

Recent studies in multiple sclerosis (MS) showed longer survival times from clinical onset than older hospital-based series. However estimated median time ranges widely, from 24 to 45 years, which makes huge difference for patients as this neurological disease mainly starts around age 20 to 40. Precise and up-to-date reference data about mortality in MS are crucial for patients and neurologists, but unavailable yet in France. Estimate survival in MS patients and compare mortality with that of the French general population. We conducted a multicenter observational study involving clinical longitudinal data from 30,413 eligible patients, linked to the national deaths register. Inclusion criteria were definite MS diagnosis and clinical onset prior to January, 1st 2009 in order to get a minimum of 1-year disease duration. After removing between-center duplicates and applying inclusion criteria, the final population comprised 27,603 MS patients (F/M sex ratio 2.5, mean age at onset 33.0 years, 85.5% relapsing onset). During the follow-up period (mean 15.2 +/- 10.3 years), 1569 deaths (5.7%) were identified; half related to MS. Death rates were significantly higher in men, patients with later clinical onset, and in progressive MS. Overall excess mortality compared with the general population was moderate (Standardized Mortality Ratio 1.48, 95% confidence interval [1.41-1.55]), but increased considerably after 20 years of disease (2.20 [2.10-2.31]). This study revealed a moderate decrease in life expectancy in MS patients, and showed that the risk of dying is strongly correlated to disease duration and disability, highlighting the need for early actions that can slow disability progression.

Zodasiran, an RNAi Therapeutic Targeting ANGPTL3, for Mixed Hyperlipidemia.

PMID: 38809174 | Indexed in PubMed

Angiopietin-like 3 (ANGPTL3) inhibits lipoprotein and endothelial lipases and hepatic uptake of triglyceride-rich lipoprotein remnants. ANGPTL3 loss-of-function carriers have lower levels of triglycerides, low-density lipoprotein (LDL) cholesterol, high-density lipoprotein (HDL) cholesterol, and non-HDL cholesterol and a lower risk of atherosclerotic cardiovascular disease than noncarriers. Zodasiran is an RNA interference (RNAi) therapy targeting expression of ANGPTL3 in the liver. We conducted a double-blind, placebo-controlled, dose-ranging phase 2b trial to evaluate the safety and efficacy of zodasiran in adults with mixed hyperlipidemia (fasting triglyceride level of 150 to 499 mg per deciliter and either an LDL cholesterol level of ≥ 70 mg per deciliter or a non-HDL cholesterol level of ≥ 100 mg per deciliter). Eligible patients were randomly assigned in a 3:1 ratio to receive subcutaneous injections of zodasiran (50, 100, or 200 mg) or placebo on day 1 and week 12 and were followed through week 36. The primary end point was the percent change in the triglyceride level from baseline to week 24. A total of 204 patients underwent randomization. At week 24, substantial mean dose-dependent decreases from baseline in ANGPTL3 levels were observed with zodasiran (difference in change vs. placebo, -54 percentage points with 50 mg, -70 percentage points with 100 mg, and -74 percentage points with 200 mg), and significant dose-dependent decreases in triglyceride levels were observed (difference in change vs. placebo, -51 percentage points, -57 percentage points, and -63 percentage points, respectively) ($P < 0.001$ for all comparisons). Other differences in change from baseline as compared with placebo included the following: for non-HDL cholesterol level, -29 percentage points with 50 mg, -29 percentage points with 100 mg, and -36 percentage points with 200 mg; for apolipoprotein B level, -19 percentage points, -15 percentage points, and -22 percentage points, respectively; and for LDL cholesterol level, -16 percentage points, -14 percentage points, and -20 percentage points, respectively. We observed a transient elevation in glycated hemoglobin levels in patients with preexisting diabetes who received the highest dose of zodasiran. In patients with mixed hyperlipidemia, zodasiran was associated with significant decreases in triglyceride levels at 24 weeks. (Funded by Arrowhead Pharmaceuticals; ARCHES-2 ClinicalTrials.gov number, NCT04832971.).

Case report of rare presentation of schistosomiasis: delayed diagnosis of genitourinary schistosomiasis in an adolescent.

PMID: 39749885 | Indexed in PubMed

Schistosomiasis is a tropical infection endemic to developing nations that can result in chronic liver damage, renal failure, infertility, and bladder cancer. Genitourinary localization is marked by dysuria, visible hematuria, and urinary obstruction. We present the case of a 17-year-old male adolescent from a rural area of Central Africa, who arrived in Italy two years prior, exhibiting hematuria and urinary symptoms. He came to our attention with a history of terminal hematuria, dysuria, and intermittent abdominal pain since the age of ten. We conducted initial blood tests, urine analyses, and ultrasonography, all yielding negative results. Cystoscopy was conducted with biopsy of an atypical bladder lesion due to the persistence of hematuria. Histopathology revealed morphological findings indicative of *Schistosoma haematobium*. The patient received praziquantel treatment and was monitored through parasitological urine analyses. Bladder schistosomiasis should be considered in children exhibiting terminal hematuria from endemic regions. Diagnosis can be established through a urinary

microbiological examination. An endoscopic evaluation may assist in the diagnosis if the results are negative.

Efficacy and safety of single-dose artesunate plus sulfalene/pyrimethamine combined with praziquantel for the treatment of children with *Schistosoma mansoni* or *Schistosoma haematobium* in western Kenya: a randomised, open-label controlled trial.

PMID: 38943214 | Indexed in PubMed

Reliance on praziquantel for the treatment and control of schistosomiasis is likely to facilitate the emergence of drug resistance. Combination therapy targeting adult and juvenile schistosome worms is urgently needed to improve praziquantel efficacy and delay the potential development of drug resistance. We assessed the efficacy and safety of single-dose praziquantel combined with single-dose artesunate plus sulfalene-pyrimethamine in the treatment of Kenyan children with schistosomiasis. This was an open-label, randomised clinical trial involving 426 school-aged children (7-15 years old) diagnosed with *Schistosoma mansoni* (by Kato-Katz) or *S. haematobium* (by urine filtration). They were randomly assigned (1:1:1) to receive a single dose of praziquantel (40 mg/kg), a single dose of artesunate plus sulfalene-pyrimethamine (12 mg/kg artesunate) or combination therapy using a single dose of praziquantel (40 mg/kg) combined with a single dose of artesunate plus sulfalene-pyrimethamine (12 mg/kg artesunate). The primary outcome was cure and egg reduction rates at 6 weeks post-treatment in the available case population. Adverse events were assessed within 3 h after treatment. Of the 426 children enrolled, 135 received praziquantel, 150 received artesunate plus sulfalene-pyrimethamine, and 141 received combination therapy. Outcome data were available for 348 (81.7%) children. For *S. mansoni*-infected children ($n = 335$), the cure rates were 75.6%, 60.7%, and 77.8%, and the egg reduction rates were 80.1%, 85.0%, and 88.4% for praziquantel, artesunate plus sulfalene-pyrimethamine, and combination therapy, respectively. For *S. haematobium*-infected children ($n = 145$), the corresponding cure rates were 81.4%, 71.1%, and 82.2%, and the egg reduction rates were 95.6%, 97.1%, and 97.7%, respectively. Seventy-one (16.7%) children reported mild-intensity adverse events. The drugs were well tolerated and no serious adverse events were reported. A single oral dose of praziquantel combined with artesunate plus sulfalene-pyrimethamine cured a high proportion of children with *S. haematobium* but did not significantly improve the treatment efficacy for either urinary or intestinal schistosomiasis. Sequential administration of praziquantel and artesunate plus sulfalene-pyrimethamine may enhance the efficacy and safety outcomes.

Pediatric urogenital schistosomiasis diagnosed in France.

PMID: 38212419 | Indexed in PubMed

Schistosomiasis affects approximately 230 million people worldwide. There is an increased incidence of schistosomiasis cases in France acquired from outside the country. This increases the risk of schistosomiasis outbreaks as observed in Corsica. Clinicians from non-endemic regions are not accustomed to diagnosing and managing this pathology. The objective of this study is to provide a better description of the clinical and paraclinical characteristics and disease evolution of affected children. Through the French Pediatric Nephrology Society and the Pediatric Infectious Pathology Group, we contacted all French pediatric centers that may have treated children with urinary schistosomiasis

between 2013 and 2019. Age, sex, comorbidities, and clinical, biological, and radiological data (at discovery and follow-up) were collected retrospectively. A total of 122 patients from 10 different centers were included. The median age was 14 years and the sex ratio M/F was 4:1. Hematuria was present in 82% of the patients while urinary tract abnormality was found in 36% of them. Fourteen patients (11%) displayed complicated forms of urinary schistosomiasis including 10 patients with chronic kidney disease. A total of 110 patients received treatment with praziquantel, which was well-tolerated and led to clinical resolution of symptoms in 98% of cases. Patients with schistosomiasis present frequent kidney, urinary, or genital involvement. Systematic screening of patients returning from endemic areas is therefore recommended, especially since treatment with antiparasitic drugs is effective and well-tolerated. Enhancing medical knowledge of this pathology among all practitioners is essential to improve care and outcomes.

Praziquantel efficacy, urinary and intestinal schistosomiasis reinfection - a systematic review.

PMID: 36394218 | Indexed in PubMed

Praziquantel (PZQ) has been extensively used as the drug of choice for the treatment of schistosomiasis on account of its safety and effectiveness against all major forms of schistosomiasis. However, low cure rate, reduced susceptibility of *Schistosoma mansoni* to PZQ and treatment failures in *S. haematobium* infections have been reported, raising concerns about its efficacy. Using the search terms, 'praziquantel efficacy, schistosomiasis, school children, reinfection' as well as defined inclusion criteria, and guided by the PRISMA guidelines, articles from 2001 to 2022 were selected from the PubMed and Google Scholar databases and reviewed to assess their importance to the research question. This review assessed the efficacy of PZQ against schistosomiasis and reinfection rates following treatment of *Schistosoma* infections in children. Majority of both intestinal and urinary schistosomiasis studies reported comparable egg reduction rates (ERRs) of 94.2% to 99.9% and 91.9% to 98%, respectively. However, ERRs suggestive of sub-optimal PZQ efficacy as well as generally high and comparable cure rates for intestinal (81.2%-99.1%) and urinary (79%-93.7%) schistosomiasis studies were reported. Schistosomiasis reinfection rates varied widely for urinary (8.1%-39.6%) and intestinal (13.9%-63.4%) studies within eight to 28 weeks following PZQ treatment. Praziquantel treatment of urinary and intestinal schistosomiasis should be accompanied by the provision of potable water, toilet, and recreational facilities to reduce reinfection and egg reduction rates and increase cure rate to expedite schistosomiasis elimination.

Botanical Products in the Treatment and Control of Schistosomiasis: Recent Studies and Distribution of Active Plant Resources According to Affected Regions.

PMID: 32823660 | Indexed in PubMed

Schistosomiasis, a parasitic disease caused by trematodes of the genus *Schistosoma*, is the second most prevalent parasitic disease in the world. It affects around 200 million people. Clinical treatment, prophylaxis, and prevention are performed in countries susceptible to schistosomiasis. In the pharmacological treatment for an acute form of schistosomiasis, the use of antiparasitics, mainly praziquantel, is more common. As an alternative way, prevention methods such as reducing the population of intermediate hosts (mollusks) with molluscicides are important in the control of this disease

by interrupting the biological cycle of this etiological parasite. Despite the importance of pharmacological agents and molluscicides, they have side effects and environmental toxicity. In addition, they can lead to the development of resistance enhancing of parasites, and lead to the search for new and effective drugs, including resources of vegetal origin, which in turn, are abundant in the affected countries. Thus, the purpose of this review is to summarize recent studies on botanical products with potential for the control of schistosomiasis, including anti-Schistosoma and molluscicide activities. In addition, species and plant derivatives according to their origin or geographical importance indicating a possible utility of local resources for countries most affected by the disease are presented.

Use of wogonin as a cooperative drug with praziquantel to better combat schistosomiasis.

PMID: 35654701 | Indexed in PubMed

Schistosomiasis is one of the most devastating tropical diseases in the world. Currently, praziquantel (PZQ) represents the best pharmacological option for the treatment of schistosomiasis as it effectively kills the worm. However, the inability to reverse established liver damages often makes treatment futile. In the current study, we investigate whether combining the use of wogonin, a compound that was found to be liver-protective, with PZQ can attribute to the greatest beneficial effect in *Schistosoma mansoni*-infected mice. To determine the protective effect of PZQ-wogonin treatment on *S. mansoni*-infected mice, histopathological analysis was done to evaluate the granuloma size and fibrotic areas in the liver. Western blotting was performed to analyze several injuries-related markers including fibrotic markers, inflammasomes, and apoptotic markers. Scanning electron microscopy was done to evaluate the effect of wogonin on the worms, and the worm and egg burden was calculated. Our results showed that PZQ-wogonin treatment significantly improved liver histopathology of *S. mansoni*-infected mice. Further analysis showed that PZQ-wogonin combinations are more effective in reducing fibrosis, inflammation, and apoptosis in the liver than that of individual drug use. Furthermore, our results revealed that wogonin is anthelmintic; and it works better with PZQ in reducing hepatic egg burden, further lessen the disease progression. In general, this combinatorial strategy may represent a new and effective approach to schistosomiasis treatment.

Three monthly doses of 60 mg/kg praziquantel for *Schistosoma haematobium* infection is a safe and effective treatment regimen.

PMID: 32375658 | Indexed in PubMed

Praziquantel (PZQ) is the standard treatment for Schistosomiasis in sub-Saharan Africa. However, there is evidence suggesting praziquantel treatment failure in Schistosome infections with associated potential renal impairment. The objective of this study was to determine the effect of three monthly doses of 60 mg/kg/day PZQ on schistosome egg count, liver and renal function during the treatment of urinary schistosomiasis in Ghana. A nested case-control study was designed from a cohort screened for schistosomiasis; 28 schistosomiasis positive cases by microscopy matched with 53 healthy controls by age and gender. The study population was urban dwellers from the Asokwa sub-metropolitan area, Kumasi in Ghana. Participants were within the age range of 6 to 30 years. We assessed *Schistosoma haematobium* egg counts in urine and its associated impact on liver and renal function at baseline,

treatment and post-treatment phases using serum. Of the 28 cases and 53 controls, 78.6% and 81.1% were males respectively. Globulin levels before treatment was higher in cases [36.7 (32.8, 40.1) vrs 30.5 (22.4, 33.8), $p = 0.005$] at pre-treatment but not at post-treatment [35.8 (31.2, 39.1) vrs 37.4 (29.7, 43.0), $p = 0.767$]. Estimated cure rate was 42.9, 46.4 and 96.4% after first, second and third dose respectively. Schistosome egg counts dropped significantly ($p = 0.001$) from before second dose to post-treatment. Similarly, levels of alanine aminotransferase ($p = 0.001$), aspartate aminotransferase ($p = 0.028$) and gamma glutamyl transferase ($p = 0.001$) significantly declined towards post-treatment. Estimated glomerular filtration rate significantly improved from before second dose to post-treatment using both the Chronic Kidney Disease Epidemiology Program ($p = 0.001$) and 4-variable Modification of Diet in Renal Disease ($p = 0.002$) equations. Treatment of urinary *Schistosoma hematobium* infections with a repeated high monthly dose of 60 mg/kg of praziquantel for 3 months is safe and effective.
